

Exploring Structure Diversity in Atomic Resolution Microscopy With Graph

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The emergence of deep learning (DL) has provided great opportunities for the high-throughput analysis of atomic-resolution micrographs. However, the DL models trained by image patches in fixed size generally lack efficiency and flexibility when processing micrographs containing diversified atomic configurations. Herein, inspired by the similarity between the atomic structures and graphs, a few-shot learning framework based on an equivariant graph neural network (EGNN) to analyze a library of atomic structures (e.g., vacancies, phases, grain boundaries, doping, etc.) is described, showing significantly promoted robustness and three orders of magnitude reduced computing parameters compared to the image-driven DL models, which is especially evident for those aggregated vacancy lines with flexible lattice distortion. Besides, the intuitiveness of graphs enables quantitative and straightforward extraction of the atomic-scale structural features in batches, thus statistically unveiling the self-assembly dynamics of vacancy lines under electron beam irradiation. A versatile model toolkit is established by integrating EGNN sub-models for single structure recognition to process images involving varied configurations in the form of a task chain, leading to the discovery of novel doping configurations with superior electrocatalytic properties for hydrogen evolution reactions. This work provides a powerful tool to explore structure diversity in a fast, accurate, and intelligent manner.

1. Introduction

Revealing the material structure at atomic resolution is the cornerstone to establishing structure-property correlation for advanced material design.^[1–4] Theoretically, the material properties are determined by the atoms and their spatial arrangements,^[5,6] which nowadays can be directly visualized by advanced transmission electron microscopy (TEM) in picometer-level precision.^[7,8] Assisted by the TEM technique, diverse atomic structures like topological defects,^[9,10] point defects,^[11–13] grain boundaries,^[14–16] and stacking configurations^[17–19] have been extensively investigated, promoting the discovery of peculiar physical properties and their potential applications in electronics, optics, and energy storage, etc.^[20–22] Essentially, an atomic-resolution TEM image is the 2D projection of a 3D structure,^[7,23] where not only are the top-view atom arrangements visualized but the composition information of each atomic column can also be reflected

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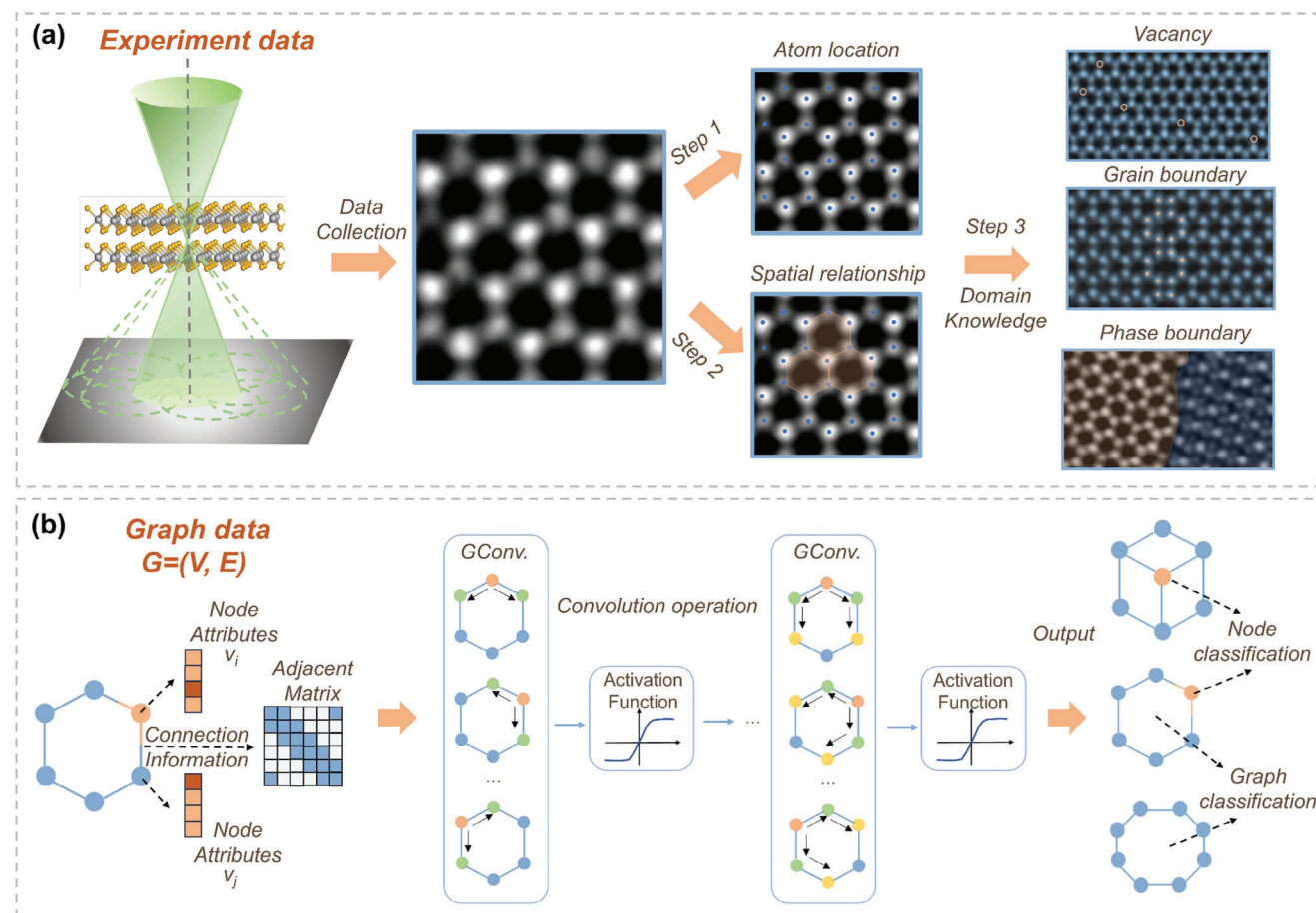


Figure 1. Schematics showing the similarity between the manual analysis process of an ADF-STEM image and the graph neural network (GNN) modeling process. a) Manual analysis process by taking an ADF-STEM image of MoS₂ as a typical example. The blue dots represent the atomic centroid after pinpointing the atom location (Step 1) and the orange semitransparent area shows the hexagonal structure of MoS₂ after determining the spatial relationship (Step 2). Different atomic structures like vacancies (orange circles), grain boundaries (orange dots), and stacking configurations (orange and blue semitransparent areas) therefore can be identified after evaluating the local gray values of each atomic column and their spatial arrangements (Step 3). b) Graph structure and the GNN modeling process. Graph data is defined as a set of nodes (V) and edges (E), where the relation between nodes is described via an adjacent matrix. GConv. refers to the graph convolution process and the black arrows represent the message-passing directions, where each graph convolution layer enables the nodes to aggregate messages and update its embedding from their nearest neighbors (from the orange node to the green node after one layer of convolution, and to the yellow node after an additional layer of convolution). After a series of graph convolution operations, different nodes or graphs can be classified according to the output feature vectors mapped by the activation function.

in some specific imaging modes, such as annular dark-field scanning transmission electron microscopy (ADF-STEM). Therefore, once a TEM image is obtained, the analyzing process by a domain expert involves the following steps: i) pinpointing each projection point; ii) determining the coordination environment based on the spatial distribution of the atomic columns; and iii) identifying the microstructure with domain knowledge after evaluating the local gray values of each atomic column and their spatial arrangements (Figure 1a). However, the detection, classification, and quantitative analysis of all the atomic structures by humans requires immense labor with biased judgment, which is particularly pronounced for high-throughput investigation. Therefore, an automated method with promoted accuracy, efficiency, and statistical rigor is highly expected.

Fortunately, DL algorithms have gradually displayed superiority in the image processing fields,^[24–26] owing to the exceptional ability to map the relation between images and contents.

Progress has been achieved in vacancy identification,^[27–29] phase segmentation,^[30–32] and crystal symmetry determination,^[33] where even massive defect recognition and statistical analysis of a low signal-to-noise ratio dataset can be realized.^[34,35] Typically, the DL algorithms can be divided into unsupervised and supervised strategies depending on whether annotated training datasets are required. Unsupervised learning supports the discovery of inherent relations among unlabeled data using clustering or association without upfront human intervention, which has been employed in the classification of relatively ordered structure motifs like vacancies and doping.^[29,36,37] However, it may encounter difficulties when dealing with microscopy datasets containing complex structures due to a lack of predefined target variables. In contrast, supervised learning can achieve superior prediction accuracy and flexibility over a wider range of complex tasks when the model is trained with sufficient manually labeled data or simulation data with ground truth.^[38–40]

However, applying TEM simulation images with labels generated by software as the training dataset is only suitable for the construction of simple defects (e.g., discrete vacancies and doping) but is difficult to batch produce complex and diversified atomic structures (e.g., grain boundaries and aggregated vacancies with flexible lattice deformation), while the use of experimental TEM images as the training sets faces the limitations of high data acquisition cost and time-consuming manual annotation.^[41,42] Furthermore, a supervised learning model that is trained by fixed-size patches generally lacks compatibility and efficiency when local regions display a range of structural variations, leading to overfitting issues and the consumption of computing sources.^[43] Therefore, it is highly desirable to develop a versatile supervised DL framework with an effective and flexible data representation approach that enables to swiftly and accurately identify various atomic structures using a limited training dataset for practical TEM image analysis.

Given the similarity between crystal structures and graphs, atomic-resolution TEM images can be effectively represented by a set of nodes and edges, (Figure 1b), where nodes are atoms, and edges describe the coordination between atoms via an adjacent matrix. During graph learning, each node aggregates messages and updates its embedding from the neighboring nodes in vector representations at every convolution layer. Iterative message passing enables each node to build representations that capture larger and more complex patterns for the classification of various atomic structures at node or graph level.^[44–46] Compared to those image-driven DL models, graph learning offers three distinctive advantages in processing atomic-resolution TEM images. First, graph representation that only contains node and connection information greatly simplifies the data structure, guiding the neural network to concentrate on atoms and their coordination manners, which are closely related to the atomic-scale structural identification, rather than wasting attention on those pixels representing the vacuum, thus outputting good inference results with less training data, time and computing power consumption. Second, the receptive field in graph learning can perfectly adapt to the structural variation according to connections between nodes, thus promoting flexibility for feature extraction against structural diversity. Finally, graph representation shows good interpretability as the model outputs well align with the atomic structure, facilitating a direct knowledge acquisition from the experimental data.

Herein, by further incorporating the geometric rotation invariance in the Euclidean plane, we build a graph learning framework based on an EGNN architecture for atomic-resolution (S)TEM investigation. With few-shot training, our framework demonstrates excellent flexibility and robustness in identifying a library of atomic structures (including vacancies, stacking configurations, doping, etc.), particularly excels in the case with flexible lattice distortion (like the aggregated vacancy lines) compared to those image-driven DL models. In addition, the intuitiveness of graphs helps quantitative and batch extraction of atomic structure features (e.g., defect concentration, local coordination manner, atomic-scale mixing state, etc.), thus statistically revealing a homophilic and anisotropic self-assembly behavior of vacancy aggregates in semiconducting MoS₂ monolayers driven by electron beam irradiation. Furthermore, we develop a versatile model

toolkit by assembling the trained EGNN sub-models for single atomic structure identification, which enables to uncover comprehensive configurations in atomic-resolution images in the form of a task chain. It uncovers a novel Pt doped on defective grain boundary structure with superior catalytic properties for hydrogen evolution reactions, demonstrating the machine learning contribution to the new structure and knowledge discovery in material science.

2. Result and Discussion

The core concept of our framework is rooted in graph learning, which is illustrated by taking the recognition process of pristine Mo, 2S (double S atoms overlapped vertically), and the defective S vacancy sites (one S missing at the 2S site) in an ADF-STEM image of monolayer MoS₂ as a study case (Figure S1, Supporting Information). To transfer raw image into graph data, in the first step, an optimized U-Net accompanied by Otsu and circular Hough transform algorithm (Figure 2a) was employed to pinpoint the atomic columns for feature point map generation (Figure 2b). The second step is planar graph generation, where we programmatically connected the identified atomic columns with their neighbors via Delaunay triangulation in a tuned edge length (Figure 2c). In the third step, graph samples were automatically extracted by searching the first or second nearest neighbors of each node from the generated planar graph, where all the nodes in the extracted graph samples were then assigned a spatial coordinate and local gray value as their initial embeddings (Figure 2d). Finally, a trained GNN was employed to aggregate the feature vector of each node from its neighbors and update the preset one into a high-dimensional representation (Figure 2e). After multi-layer aggregation, the high-level feature vector of each graph sample was then input into the Softmax classifier to obtain the corresponding probability for classification at node or graph level (Figure 2f).

In the first step of our workflow, a multi-scale U-Net^[47] was applied to simultaneously extract features at multiple resolutions through downsampling the intermediate feature maps of the input ADF-STEM image, providing explicit guidance from distinct prediction branches to enhance the performance of the vanilla model (Figure S2, Supporting Information). The as-generated probability map was subsequently binarized as foreground and background regions by an Otsu algorithm,^[48] which can adaptively optimize the segmentation threshold by maximizing the between-class variance (Figure S3, Supporting Information) in terms of feature distribution variation between the probability maps. Finally, a circular Hough transform algorithm^[49] was employed to transform the as-detected foreground into a circular object, and the circle center was thereafter identified as the centroid of the atomic columns. The optimized algorithm enables pinpointing atomic columns with an F1 score of 99.8%, even for the images suffering from astigmatism or surface contamination (Figure S4, Supporting Information), laying a good foundation for the subsequent graph construction.

A general connection method based on Delaunay triangulation was then implanted to locate the neighbors of each feature point and create edges for planar graph generation, with which the graph samples can be directly extracted via local

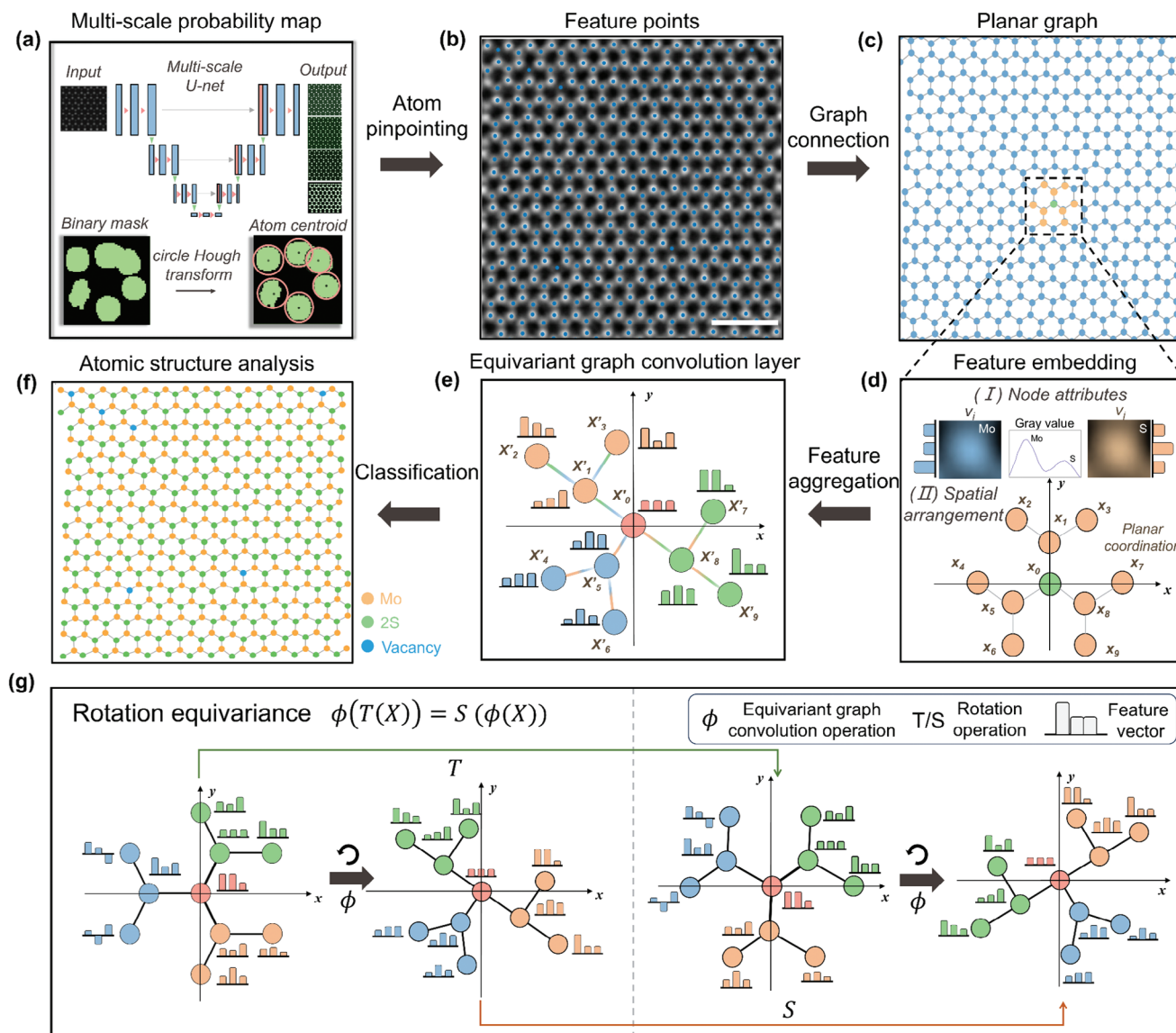


Figure 2. DL workflow to identify atomic structures by taking an ADF-STEM image of monolayer MoS₂ containing vacancies as a typical example. a) Illustration of the atom pinpointing architecture. The top half shows a multi-scale U-Net analysis process that generates predictions from multiple resolutions. The bottom half shows the visual results of the output probability map from the multi-scale U-Net after experiencing binarization via the Otsu algorithm and centroid detection via the circular Hough transform algorithm sequentially. b) ADF-STEM image of monolayer MoS₂ with pinpointed feature points. Scale bar: 1 nm. c) Programmatic graph generation via Delaunay triangulation connection. d) Graph samples extracted by searching for the first- and second-nearest neighbors with any detected node as the graph center (green dot) for initial feature embedding. The gray value of each atomic column is embedded in its corresponding node in the form of a feature vector. The planar coordination of each node is extracted by taking the graph center as the origin. e) Schematic exhibiting the output graph structure after feature aggregation via the trained EGNN, where the initial feature embedding of each node has been updated by aggregating their neighbors' features. f) Identification results corresponding to the atomic sites of Mo, 2S, and S vacancies in the form of the planar graph from b). g) Schematic showing the rotation equivariance of EGNN, which can be described as an operation where the output graph feature embedding (the second panel) synchronously changes with the input graph feature embedding (the first panel) after experiencing rotation operation ("T" corresponds to the rotation operation of the input graph embedding and "S" corresponds to the rotation operation of the output graph embedding).

neighborhood search in a universal setting. Note that, to avoid disconnection or overcounting the distant neighbors during the edge generation process, we set a length threshold for the Delaunay edge according to the lattice parameters, where any edges that exceed this length were removed, ensuring the planar graph to be maximally consistent with the crystal structure (Figure 2c;

Figure S5, Supporting Information). Once graph samples have been extracted, we assigned every node a flattened 9D feature based on the local gray value of the corresponding atomic column, which contains the composition information (Figure 2d; Figure S6, Supporting Information), and a planar coordinate to describe its spatial arrangements.

Since rotation transformation will not break the geometry symmetry but change the graph representation in the vector space, the graph feature learning may encounter difficulties when presented with structural variations significantly different from the training data. Therefore, a geometric equivariance with respect to the rotation transformations in the Euclidean plane is highly desired to enhance the robustness of the as-learned graph features. Given that, in the final step of our framework, we employed an emerging model named EGNN^[50] to extract graph features under the equivariant restriction against rotation variation. Different from vanilla GNN models, the graph convolution layer in our EGNN model is designed to aggregate the spatial position information of each node during the edge operation process, enabling a simultaneous feature update of embedded values and spatial position against rotation transformation to make the output graph representation more robust (Figure 2g; Figure S7, Supporting Information).

This framework readily extends to identify boundaries and interfaces across various materials involving different compositions, dimensionalities, lattice symmetries, and functionalities (Figures S8 and S9, Supporting Information). Examples include 2H-3R stacking boundary and grain boundary identification of atomically thin MoS₂ (Figure 3a(i),(ii)), phase boundary segmentation (between O3 and rock salt phase) in a layered cathode lithium-ion battery material of NCM 811 (referring to LiNi_{0.8}Co_{0.1}Mn_{0.1}O₂) (Figure 3a(iii)), and hetero-interface recognition between bulk elemental materials of aluminum (Al) and silicon (Si) (Figure 3a(iv)). We take the ADF-STEM image segmentation of 2H, 3R, and 1H stacking configurations in MoS₂ as a typical study case (Figure 3b). By searching the first nearest neighbors of each identified node from the generated planar graph (Figure S10, Supporting Information), graph samples corresponding to different stacking configurations (Figure 3c) can be automatically extracted despite the structure diversity, which aligns well with their crystal structures, indicating the flexibility and intuitiveness of Delaunay triangulation for the graph generation. With few-shot training (each configuration with ≈ 200 graph samples extracted from two pieces of ADF-STEM image, Figure S11, Supporting Information), the EGNN model attains a steady decrease in loss (obtained by the cross entropy loss function, see details in Experimental Section) and shows a minor mean squared error (MSE) of 0.019 between the training and validation loss values (Figure 3d), proving the validity of EGNN model for few-shot learning in graphs. Although the loss obtained by the CNN model (ResNet 18)^[51] also exhibits a rapid convergence trend after training several epochs, a large gap between the validation and training loss values is observed with an MSE of 0.341, indicating a pronounced overfitting issue. It may be ascribed to the excessive redundant information extracted from image patches in such a limited training dataset. As a result, the EGNN model achieves an impressive F1 score of over 0.98 for the classification of stacking configurations in this test case (Figure S13a, Supporting Information), which outperforms the CNN model showing an F1 score of 0.60 (Figure S12, Supporting Information). More importantly, nearly 8000 times reduced computing parameters are performed via graph learning, displaying 3.9 times promoted inference efficiency compared to that using the image-driven model (Figure 3e; Table S1, Supporting Information). Figure 3f shows the end-to-end visualization results ob-

tained by our framework, whereby a clear boundary can be identified between different stacking configurations (Figure S13b, Supporting Information). Note that, all the misidentifications are primarily situated at the stacking boundary, which are inherently indistinguishable even for a domain expert.

In addition to the relatively ordered microstructures studied above, our framework verifies superiority in analyzing defective structures with prominent and flexible lattice distortion. A typical example is the recognition of aggregated S vacancy lines in monolayer MoS₂, which are in-situ produced by the electron beam irradiation during the high-resolution TEM (HR-TEM) imaging process, whose structures would evolve and diversify as the irradiation time prolongs (Figure S14, Supporting Information). Distinct from the discrete S vacancies, the aggregated S vacancy lines exhibit significant geometry variations in both length and width, inducing flexible and diversified lattice distortion as large as 20.8% along the armchair direction (Figure S15, Supporting Information), which poses challenges for conventional computer vision algorithms to effectively extract features from convolution kernel or other representations in fixed patches (Figure S16, Supporting Information). On the contrary, graph representation can fix this limitation as the receptive field can well adapt to the structural variation once a connection between each node has been set. More importantly, the cognitive ability of the EGNN model in the spatial position of each node enables to detect the aggregation status of vacancies, which well aligns with the human cognitive criteria and endows the algorithm with interpretability and intuitiveness.

Likewise, we trained an EGNN model using only one HR-TEM image of the 0 s irradiation sample (image size: 2048 $\times$ 2048) and tested it on the 10 s one (Figure 4a), which has already experienced severe lattice disorder and geometry variation (Figure 4b). By searching the second-order neighbors, graph samples corresponding to different defect configurations (typical examples shown in Figure 4d) can be flexibly extracted in the as-generated planar graph (Figure 4c) regardless of the image patch size. Therefore, graph samples in varied geometry can be well separated into three clusters with a distinctive boundary in the feature space (Figure 4e), showing the highest F1 scores of 97.6% and 97.5% for the classification of aggregated line and discrete point defects, respectively (Figure 4f; Figure S18, Supporting Information). Notably, removing the spatial coordinates of each node will eliminate the model's ability to perceive the aggregate state of the vacancies, thus resulting in an increased proportion of misidentifying discrete point vacancies as the linearly aggregated ones (Figure 4f; Figure S17a, Supporting Information). Other state-of-the-art architectures like ResNet 18 and Zernike polynomial (ZP)^[37] also moderately underperform in this task (Figure S17b,c, Supporting Information), where the F1 scores for line and point defect recognition using these two models decrease to 92.5%, 84.7%, and 85.7%, 51.1%, respectively (Figure 4f; Figure S19, Supporting Information).

To further manifest the robustness of our frameworks, a series of in-situ HR-TEM images suffered from varied lattice distortion was employed for end-to-end evaluation. The recognition ability of our framework remains intact regardless of the increased lattice distortion (Figure 4g; Figure S20, Supporting Information), realizing F1 scores of 98.3% and 87.3% for the recognition of linearly aggregated and discrete vacancies in the 40 s irradiation

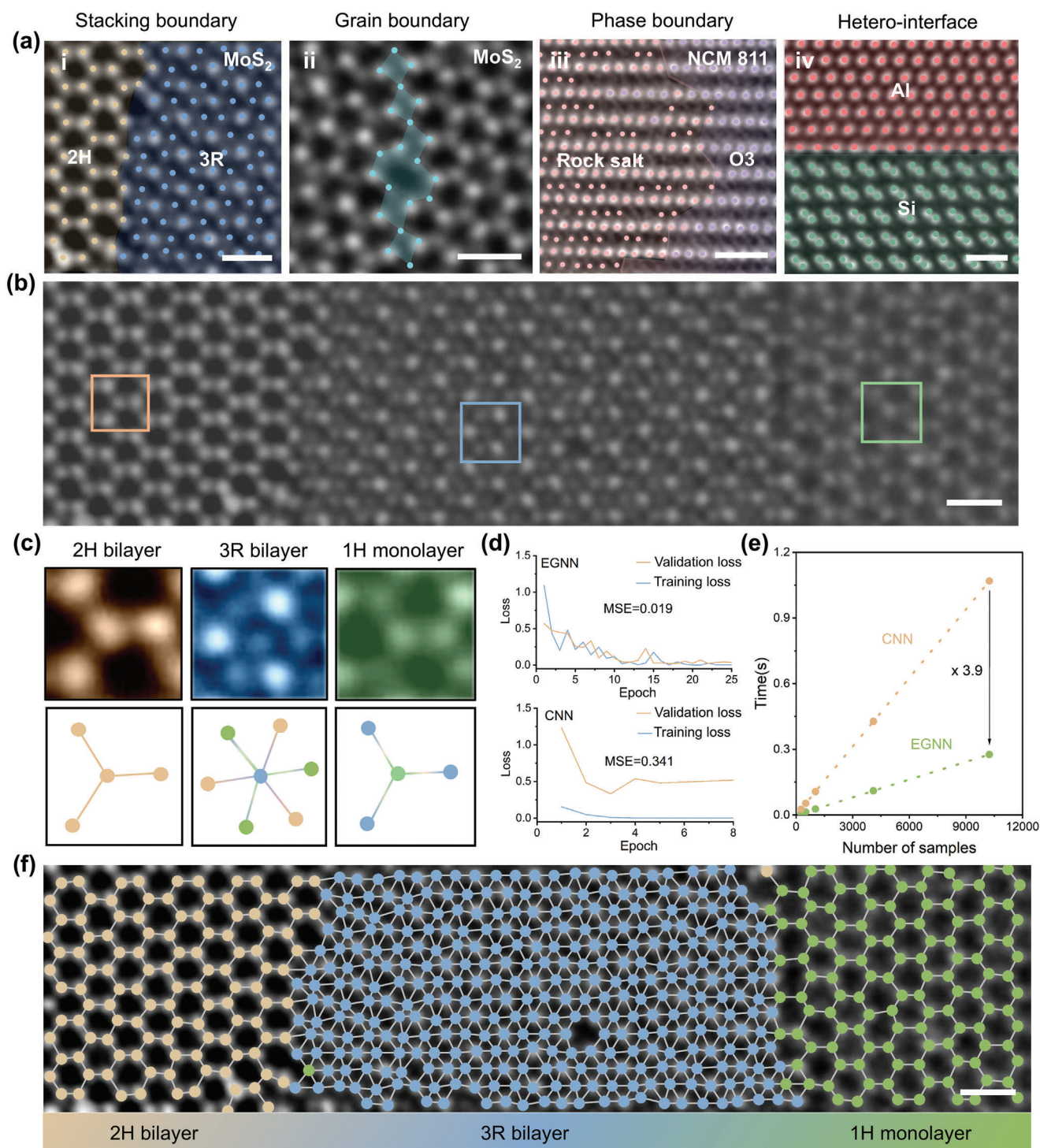


Figure 3. Identification of boundaries and interfaces across various material systems. a) Visualization results of the 2H-3R stacking segmentation in bilayer MoS_2 i), grain boundary extraction in monolayer MoS_2 ii), phase boundary segmentation of NCM811 iii), and interface identification of Al/Si heterostructure iv), respectively, inferred by our DL algorithm. The orange, blue, cyan, pink, purple, red, and green dots represent the identified atomic columns corresponding to the 2H stacking, 3R stacking, grain boundary, rock salt phase, O_3 phase, Al, and Si. Scale bars: 0.5 nm. b) ADF-STEM image of a bilayer-monolayer MoS_2 lateral heterostructure, which involves 2H (bilayer), 3R (bilayer), and 1H (monolayer) stacking configurations from left to right. Scale bar: 0.5 nm. c) Image patches of different stacking configurations from b). The orange, blue, and green nodes represent the Mo+2S, 2S, and Mo atomic sites, respectively. d) Training and validation loss generated by EGNN and CNN models. e) Plot showing the inference time that varies with the image patch or graph sample numbers using CNN and EGNN models. f) Visualization result from b) by our framework. Scale bar: 0.5 nm.

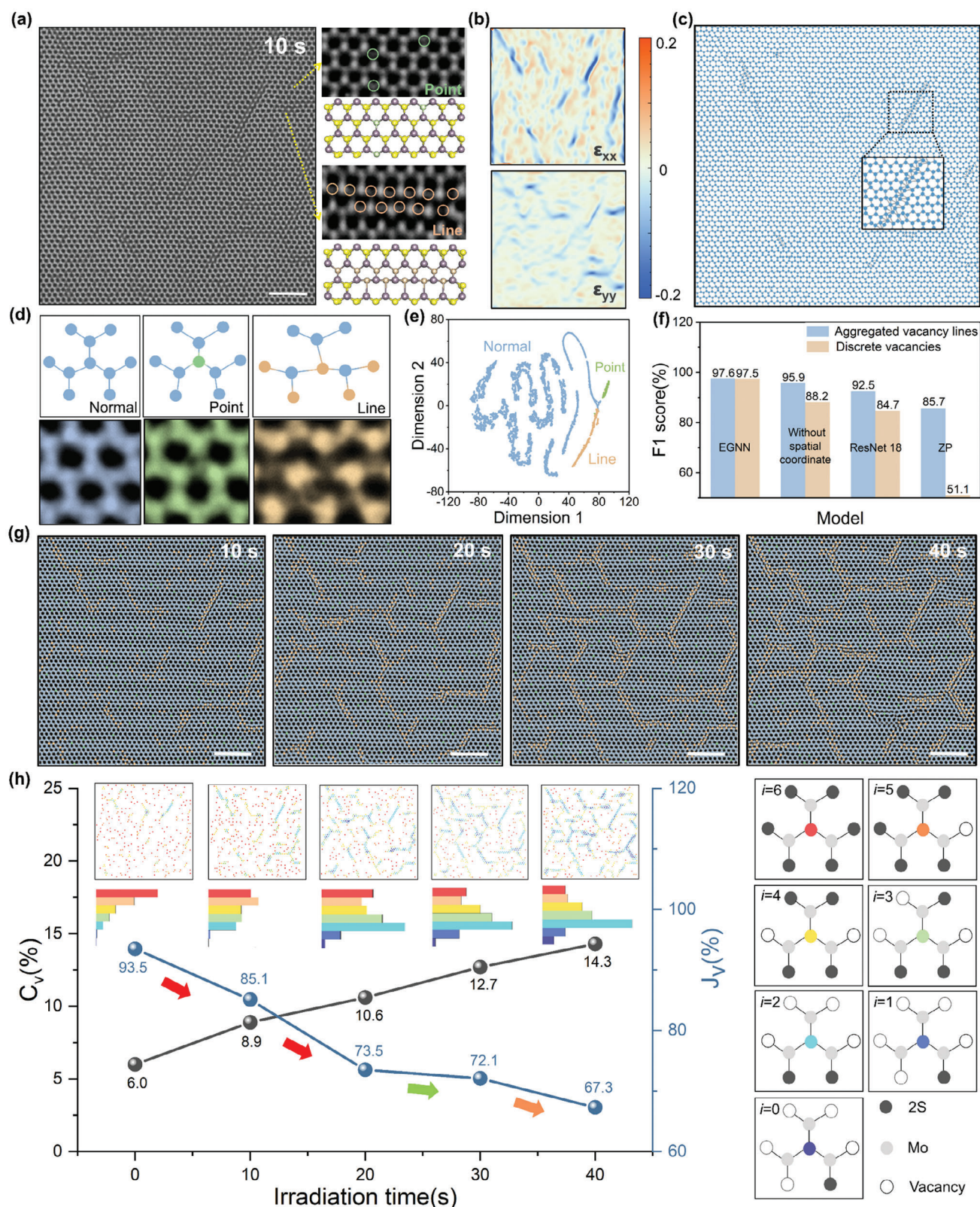


Figure 4. Automatic analysis of aggregated S vacancy lines with flexible and varied lattice distortion. **a)** HR-TEM image of monolayer MoS₂ after 10 s e-beam irradiation. The right panels present the enlarged views of the discrete S vacancies and aggregated S vacancy lines with their corresponding atomic models from top views. Scale bar: 2 nm. **b)** Strain map of **a)** in the horizontal and vertical directions through geometric phase analysis. The color scale ranges from -0.2 to 0.2. Obvious compression strain (blue area) can be observed around the vacancy lines. **c)** Planar graph representation

sample when the proportion of aggregated vacancies (ratio between aggregated vacancies and the total S atom number) has already increased from 3.5% (corresponding to the training data) to 14.3%. However, multi-scale U-Net alone fails to work in this situation (Figure S21, Supporting Information), and so do the other classic end-to-end DL models like YOLO based on object detection^[52] (Figure S22, Supporting Information). Note that, even though the number of graph samples in this task increases to more than 5000, end-to-end computing of one image by our framework only takes 0.56 s, showing a 5.5 times improvement in efficiency compared to that by CNN (Table S2, Supporting Information), realizing the real-time TEM investigation for dynamic structural evolution (Video S1, Supporting Information).

Due to the inherent intuitiveness of graphs, our framework can directly and quantitatively extract structure-related parameters (e.g., defect concentration, local coordination number, atomic-scale mixing state, etc.) from the HR-TEM image series in Figure 4g involving $\approx 27\,000$ atoms, thus statistically revealing self-assembly behavior of S vacancy aggregates in monolayer MoS₂ driven by electron beam irradiation, which may help precise defect engineering and electronic band modulation at the atomic level for advanced nanodevice design.^[53] Generally, the vacancy concentration shows a linear increase with the irradiation time from 6% in the initial state to nearly 15% after undergoing 40 s irradiation (black line in Figure 4h). A statistical analysis of the coordination number of S vacancies was then achieved by quantifying the number of 2S atomic columns located at the second nearest neighboring sites of each S vacancy (denoted as i , ranging from 0 to 6, right schematics in Figure 4h), showing a decrease of the dominant i from 6 (red columns) to 2 (cyan columns) as the S vacancy proliferate (inset bar charts of Figure 4h). It indicates vacancy aggregation and an increasingly significant deviation of the local stoichiometric ratio of Mo/S from 1:2. We further investigate the irradiation time-dependent mixing state evolution between S vacancies and the pristine 2S sites at the atomic scale by introducing a parameter called “alloying degree” (J_v).^[54] All information needed for J_v calculation can be obtained from the model outputs (Figure S23, Supporting Information). Two interesting phenomena were unclosed (blue line in Figure 4h): i) J_v is always less than 100% and continuously decreases with the irradiation time, indicating in increasingly pronounced homophilic assembly manner of S vacancies (i.e., vacancies prefer to aggregate with vacancies rather than 2S sites). ii) The decline rate of J_v displays three stages, indicating the self-assembly process of vacancy lines from nucleation to anisotropic growth (Figure S23, Supporting Information). The rapid decline stage (red arrows) corresponds to the aggregation of

isolated vacancies to form short and narrow baby vacancy lines. Then a platform stage (green arrow) follows corresponding to the extension of vacancy lines along the zigzag lattice directions in MoS₂. Finally, a moderate J_v decline stage (orange arrow) comes corresponding to the broadening of the as-grown long and narrow vacancy lines along the armchair lattice directions. Not that, only minor differences are seen between the statistical results obtained by our ML framework and those obtained by human experts (Figure S24, Supporting Information), demonstrating that our algorithm achieves accuracy comparable to that of human experts.

In practical situations, materials encompass not just a single type of defect but a library of defective configurations.^[55] However, relying on a specific model fails a comprehensive analysis of complex systems, and developing a new model is not always feasible due to the scarcity of training data. Herein, we provide a model toolkit by assembling the trained sub-models to support an all-around exploration of diverse atomic configurations in the form of a task chain. Proof of this concept is exemplified by analyzing an ADF-STEM image of Pt-doped defective MoS₂ that coexists with a range of atomic configurations such as doping, grain boundaries, stacking configurations, and vacancies (Figure 5a), where there is generally no training dataset available for such a specific level of structural complexity. To screen out all the atomic configurations, in the first step, we additionally trained a doping configuration corresponded EGNN sub-model and assembled it with the previously trained ones (including the EGNN models for identifying vacancies, grain boundaries, and stacking configurations, etc.) to establish a versatile model toolkit. Second, we transferred the input ADF-STEM image into a graph representation for feature extraction. Finally, a task chain was built by combining EGNN sub-models corresponding to identifying doping, stacking configurations, grain boundaries, and vacancies in sequence, with which the diversified atomic structures can be comprehensively analyzed by overlaying the recognition results from different models.

This process can be extended to batch analyze a series of data involving 10^4 atoms, which is expected to discover novel configurations from the models’ outputs by investigating the combinations among the “known” ones (e.g., if different EGNN sub-models recognize the same location in a micrograph for the existence of corresponding microstructures, it indicates the potential emergence of a new “unknown” configuration here that is constructed by the combination of these submodel-corresponded “known” single structures). A detailed investigation of the Pt doping sites and their local atomic environments was realized by stepwise overlaying and analyzing the identification results of

generated from a). d) Typical graph samples centered on different types of atomic columns with their corresponding image patches from a) to illustrate the flexibility of graph representation. The blue, green, and orange dots represent the normal atomic columns, discrete vacancies, and linearly aggregated vacancies, respectively. e) T-SNE outputs in feature space generated by the trained EGNN model from a). f) Performance metrics of EGNN model and other representative algorithms tested on a) for comparison. g) End-to-end recognition results by our framework tested on the HR-TEM image series of monolayer MoS₂ after experiencing 10, 20, 30, and 40 s e-beam irradiation. Scale bars: 2 nm. h) Plots showing the vacancy concentration (C_v) and the alloying degree of S vacancies (J_v) as a function of irradiation time. The inset images show the quantity variation of vacancies that are paired with 6 (red), 5 (orange), 4 (yellow), 3 (green), 2 (cyan), 1 (blue), and 0 (dark blue) 2S atomic columns as the irradiation time prolongs, which can be directly extracted from the graph outputs by counting the number of disulfide atomic columns on the second nearest neighboring sites of each S vacancy (illustrated in the right panels, i is the coordination number ranging from 0–6). The inset bar charts show the corresponding ratio variation of the vacancies paired with different numbers of 2S atomic columns. The red, green, and orange arrows illustrate the rapid decline stage, the platform stage, and the moderate decline stage, respectively.

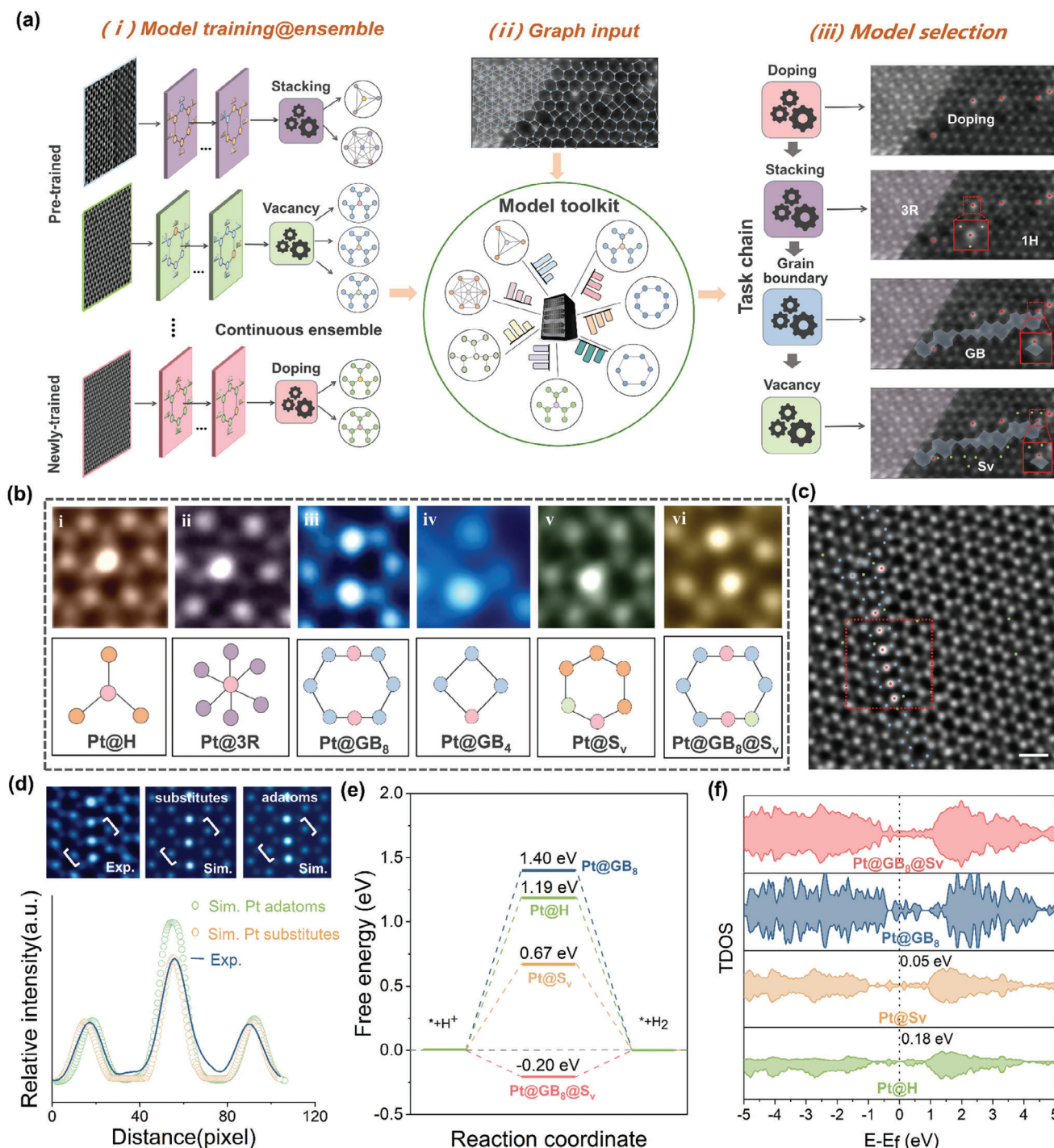


Figure 5. Comprehensively exploring diversified atomic structures by integrating different sub-models in the form of a task chain. a) Schematic showing the sequential analysis process of various types of atomic configurations in the form of a task chain by employing different EGNN sub-models. An ADF-STEM image of Pt-doped MoS₂ containing a range of atomic configurations such as doping, grain boundaries, stacking configurations, and vacancies is exemplified here. The red squares in the right panels show the enlarged views of typical Pt doping configurations after stepwise overlaying the recognition results from different models' outputs. b) Extracted patches of different Pt doping configurations with their corresponding graphs. The pink, orange, purple, blue, and green nodes represent Pt, 1H monolayer, 3R stacking, grain boundaries, and vacancies, respectively. c) ADF-STEM image showing the Pt atom situated at the 8-fold rings of Mo sites in the form of an aggregated row and its corresponding identification results. The red, blue, and green dots represent the doping, grain boundaries, and vacancies. d) Intensity line profiles measured between the experiment and simulation images. The experimental image is extracted from the red square marked in d). We used two simulation models, Pt substitutes of Mo and Pt adatoms on Mo around the 8-fold rings of grain boundary, to compare their intensity line profiles with the experimental data, where the intensity line profile corresponding to the Pt substitutes of Mo fits better with the experiment data. e) Calculated hydrogen adsorption free energy for different Pt doping configurations. f) TDOS plots calculated from different Pt doping configurations.

Pt doping with that of the others (stacking, grain boundary, and vacancy corresponded models, see details in Figure S25, Supporting Information). Beyond the extensively investigated doping configurations^[56] like the Pt located at the pristine 1H monolayer (Pt@1H) and 3R-stacked bilayer MoS₂ (Pt@3R) with no defect around (Figure 5b(i),(ii)), our framework discovered a larger range of doping diversity: Pt located at the 8-fold rings of grain boundaries (Pt@GB₈, Figure 5b(iii)), Pt located at the 4-fold rings of grain boundaries (Pt@GB₄, Figure 5b(iv)), Pt located at the basal plane of 1H monolayer and at 8-fold rings of grain boundaries with S vacancy around (Pt@S_v in Figure 5b(v), Pt@GB₈@S_v in Figure 5b(vi)). In addition, when tracing the distribution of different Pt@GB configurations (Figure S25, Supporting Information), it can be found that Pt atoms are prone to locate at the 8-fold rings of grain boundaries and an aggregated Pt row along the grain boundary with vacancy around are detected (Figure 5c). By further analyzing the intensity line profiles between the simulation and the experimental data, these Pt atoms are verified to be the substitutes of Mo rather than the adatoms (Figure 5d).

Since Pt is highly regarded as a top-performing dopant for promoting the electrocatalytic performance of 2D MoS₂ in hydrogen evolution reaction (HER),^[2] the discovery of these novel doping configurations motivates us to investigate their potential catalytic activity for hydrogen production. DFT calculations were conducted to examine the adsorption free energy of H* (ΔG_H) of Pt@GB₈ and Pt@GB₈@S_v. Pt@S_v and Pt@1H (Pt situated in the metal site) were employed for performance comparisons, where 2S or S_v sites around the Pt atoms were chosen as the adsorption sites according to the previous works.^[57] (Figure S26, Supporting Information). Figure 5e shows that both Pt@GB₈@S_v and Pt@S_v exhibit superior ΔG_H of -0.20 and 0.67 eV than that of Pt@GB₈ (1.40 eV) and Pt@1H (1.19 eV), indicating a dominant role of the vacancy-Pt pair in promoting the catalytic properties. Note that, the vacancy-Pt pair situated in the 8-fold rings of grain boundaries pushes ΔG_H close to 0 eV, which is more beneficial to compromise the reaction barriers in the adsorption and desorption steps.^[58] Electrical conductivity is another key factor for catalytic activity. Good electrical conductivity ensures efficient electron transfer during the HER process. Figure 5f shows the total density of states (TDOS) of these four doping configurations. Both Pt@GB₈ and Pt@GB₈@S_v introduce electronic states around the Fermi level, showing a semiconductor-to-metal transition. Although the position of the valence band moves downward when forming a vacancy-Pt pair in the Pt@1H, the Pt@S_v configuration still exhibits a semiconductor property with a minor band gap of 0.05 eV. The above results offer valuable insights into advanced electrocatalyst design through doping configuration modulation, showcasing our framework's contribution to new knowledge emergence.

3. Conclusion

To sum up, we describe a few-shot learning framework based on EGNN to investigate a series of complex atomic resolution micrographs (Table S3, Supporting Information). The focus of the graph neural network, nodes and edges, perfectly matches key information to understanding atomic-scale crystal structures, i.e., atoms and their coordination with neighbors, avoiding wasting attention on the vacuum regions (e.g., black areas in mono-

layer MoS₂ STEM pictures) that occupy a large proportion of the microscopy images. Therefore, our framework provides flexibility, robustness, and computing efficiency in identifying various atomic configurations (vacancies, phases, grain boundaries, doping, etc.) compared to image-driven DL architectures, and performs especially well for defects with flexible and varied lattice distortion, like the aggregated vacancy lines in MoS₂. The intuitiveness of graphs enables quantitative and straightforward extraction of structure-related parameters in batches (e.g., defect concentration, local coordination number, atomic-scale mixing state, etc.), thus revealing the evolution dynamics of vacancy lines under electron beam illumination, which has potential applications in advanced nanodevice design. We also built an extendable toolkit to explore structure diversity in the form of a task chain, contributing to the discovery of novel doping configurations with high catalytic activity. This work demonstrates a computationally efficient and versatile platform that enables automatic, accurate, fast, and batch identification of diversified atomic-scale structures with a small training dataset. It is in line with the needs of practical microscopy data analysis, helps unveil statistically grounded information, and facilitates new structure and knowledge discovery.

4. Experimental Section

Growth and transfer of monolayer MoS₂: MoS₂ monolayers were synthesized by atmospheric pressure chemical vapor deposition (APCVD) without hydrogen introduction, according to previous reports.^[59] Molybdenum trioxide (MoO₃, $\geq 99.5\%$, Sigma-Aldrich), sulphur (S, $\geq 99.5\%$, Sigma-Aldrich), and SiO₂/Si (300 nm thick SiO₂) chips were used as precursors and substrates, respectively. To avoid the quench of MoO₃ powder by S vapor during the reaction, an inner tube with MoO₃ powder placed inside was inserted into the outer 1-inch quartz tube, where S powder was positioned upstream. The typical heating temperatures for S, MoO₃, and SiO₂/Si substrate were ≈ 180 , ≈ 300 , and ≈ 800 °C, respectively. The growth lasted 20 min followed by a fast cooling.

For the transfer of monolayer MoS₂, we first spin-coated a thin film of poly (methyl methacrylate) (PMMA) on the MoS₂/SiO₂/Si substrate and then floated it on a 2 mol L^{-1} potassium hydroxide (KOH) solution to peel off the PMMA/MoS₂ film from the SiO₂/Si substrate. Subsequently, the film was transferred to the deionized water three times to remove etchant residuals thoroughly. The film was scooped up by the SiN_x TEM grid, naturally dried in the air, and baked on the hotplate at 180 °C for 15 min. Finally, the PMMA scaffold was removed by submerging the TEM grid in acetone for 8 h.

(S)TEM Characterization: The ADF-STEM images of MoS₂ were captured using an aberration-corrected JEOL ARM300CF equipped with a JEOL ETA corrector at an accelerating voltage of 60 kV. Dwell times of $5\text{--}20 \mu\text{s}$ and a pixel size of 0.006 nm px^{-1} were applied. Optical conditions used a condensed lens aperture of $30 \mu\text{m}$, a convergence semi-angle of 31.5 mrad , a beam current of 44 pA , and inner–outer acquisition angles of $49.5\text{--}198 \text{ mrad}$. The HR-TEM images were conducted using an FEI Titan 80–300 environmental TEM under 80 kV accelerating voltage with a Gatan OneView (4kx4k) high frame rate camera.

Data Smoothing and Dataset Division: All the raw (S)TEM images were smoothed using a Gaussian blur method within ImageJ software, where the standard deviation (σ) of the Gaussian function was set to 3. For the atom pinpointing task, 10 pieces of labeled (S)TEM images in different sizes were employed as the training dataset, while 3 pieces of (S)TEM images were reserved for validation. The rest of the images involved in this study were set as the test dataset. For EGNN model training in each specific task, we only took one piece of labeled (S)TEM image as the training dataset and one for validation, with the remaining images assigned as the test dataset.

Multi-Scale U-Net: U-Net was a convolutional neural network architecture designed for image segmentation, which features a symmetric U-shaped structure to enable precise localization and contextual information extraction. Optimized from vanilla U-Net architecture, multi-scale U-Net was a variant of U-Net that can better capture multi-scale image features by simultaneously making predictions from different resolutions. Given an image of $p \in \mathbb{R}^{H \times W \times C}$, we use the multi-scale U-Net ϕ_{msunet} to extract multi-scale features:

$$V = \phi_{msunet}(p) = \{v_0, v_1, v_2, v_3\} \quad (1)$$

where $v_0 \in \mathbb{R}^{H \times W}$, $v_1 \in \mathbb{R}^{\frac{H}{2} \times \frac{W}{2}}$, $v_2 \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4}}$, $v_3 \in \mathbb{R}^{\frac{H}{8} \times \frac{W}{8}}$ correspond to the probability maps predicted at four scales of the multi-scale U-Net. To enhance the model's generalization and robustness, we employed various data augmentation techniques, including random horizontal and vertical flipping, gaussian blur, brightness, contrast adjustments, as well as random affine transformations such as translation, scaling, and rotation. During inference, we only took the probability map v_0 and applied the Otsu algorithm to separate the foreground and background. Then, we employed the circle Hough transform (CHT) for atomic center point extraction, obtaining a set of atomic center coordinates x^0 :

$$x^0 = CHT(OTSU(v_0)) = \{x_0^0, \dots, x_{K-1}^0\}, x_k^0 \in \mathbb{R}^2 \quad (2)$$

where K denotes the number of identified atoms, and x_k^0 represents the 2D spatial coordinates of each atom.

Equivariant Graph Neural Network: EGNN extends traditional graph neural networks by incorporating equivariance properties, ensuring that the model's outputs are invariant or equivariant to the rotation transformation of the input data. For any atom as the center, we constructed the corresponding graph data $G = (\mathcal{V}, \mathcal{E})$, where $\mathcal{E}$, $\mathcal{V}$ represents the adjacency matrix and node information, respectively. The node information $\mathcal{V}$ comprises the coordinates x and features h , which are derived from the spatial positions corresponding to the cropped and expanded images.

For each graph sample $G = (\mathcal{V}, \mathcal{E})$, the equivariant graph neural network requires three inputs, including the adjacency matrix $\mathcal{E}$, the atomic feature vector $h^l = \{h_0^l, \dots, h_{M-1}^l\}$ and the spatial coordinates of atoms $x^l = \{x_0^l, \dots, x_{M-1}^l\}$, where M denotes the number of atoms in the sample, and $l \in [0, L-1]$ denotes the layer of the equilateral graph neural network. Equivariant Graph Convolutional Layer (EGCL) takes h^l , x^l , and the edge connectivity $\mathcal{E} = (e_{ij})$ as input and fuses the information to obtain $l+1$ layer features: h^{l+1} and x^{l+1} . In short, h^{l+1} , $x^{l+1} = EGCL(h^l, x^l, \mathcal{E})$. The intra-layer operations are as follows:

$$m_{ij} = \&\phi_e \left(h_i^l, h_j^l, \|x_i^l - x_j^l\|^2 \right) \quad (3)$$

$$x_i^{l+1} = x_i^l + \frac{1}{M-1} \sum_{j \neq i} (x_i^l - x_j^l) \phi_x(m_{ij}) \quad (4)$$

$$m_i = \sum_{j \neq i} m_{ij} \quad (5)$$

$$h_i^{l+1} = \phi_h(h_i^l, m_i) \quad (6)$$

where ϕ_e and ϕ_h are the edge and node operations respectively. $\phi_x: \mathbb{R}^{df} \rightarrow \mathbb{R}^1$ takes the edge embedding m_{ij} from the previous edge operation and outputs a scalar value. According to the different tasks, there are two classification methods: the first is classification based on the central atom:

$$\gamma = \text{Softmax}(h_0^L), \gamma \in \mathbb{R}^c \quad (7)$$

where c represents the number of classes. Another approach is to perform classification on the graph structure:

$$\gamma = \text{Softmax}(\text{AGG}(h^L)), \gamma \in \mathbb{R}^c \quad (8)$$

where AGG represents the aggregation operation.

Model Assembling and Task Chain Building: In the model database assembling scenario, let's consider having T sub-models $\{\phi_1, \phi_2, \dots, \phi_T\}$, where each model ϕ_i is focused on recognizing a specific atomic type or a particular defect. Given input data $D = (G, S)$, comprising graph data G and a sequence of prediction tasks S , the input data D into the sub-models based on the sequence S was feed. With the dependency relationship defined by S among the models, each sub-model could receive outputs from other models as its input. This facilitates information exchange between models, allowing for the subdivision of diversified structure configurations.

Loss Function: Our framework consists of two parts: multi-scale U-Net semantic segmentation and EGNN defect classification. In the multi-scale U-Net semantic segmentation part, we used a combination of Dice loss and Cross entropy loss functions. Dice Loss was used to measure the overlap between predicted segmentation results and ground truth, particularly suitable for imbalanced categories; Cross entropy loss was used to measure the difference between model outputs and ground truth labels in classification tasks. The calculation formula is:

$$\text{Dice Loss} = 1 - \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \quad (9)$$

$$\text{Cross entropy Loss} = - \sum_{i=1}^N y_i \cdot \log(\hat{y}_i) \quad (10)$$

where TP , FP , and FN are true positives, false positives, and false negatives, respectively. $\hat{y}_i$ is the predicted probability of the model for the i -th sample, and y_i is the corresponding true label. In the defect classification stage of EGNN, it was only used the cross-entropy loss function to optimize category prediction results.

Evaluation Metrics: The F1 score was used, precision, and recall to evaluate the model's recognition and classification performance. Precision refers to the proportion of samples predicted as positive by the model that are actually positive. It measures the accuracy of the model in predicting positive samples. Recall refers to the proportion of actual positive samples that were correctly predicted as positive. It measures the model's ability to identify positive samples. The F1 score was the harmonic mean of precision and recall, used to balance the relationship between the two. It was particularly suitable for situations where sample categories are imbalanced. The formula is as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (11)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (12)$$

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (13)$$

where TP (True Positives) represents the number of samples correctly predicted as positive class, FP (False Positives) represents the number of samples incorrectly predicted as positive class, and FN (False Negatives) represents the actual positive class samples incorrectly predicted as negative class.

Hyperparameters: During the training process, this code was implemented based on the PyTorch Lightning framework. The EGNN model involved two EGNN layers and had an input/hidden node dimension of 9/6, respectively. The output node dimension was determined by the specific task. The AdamW optimizer for parameter optimization with a batch size of 32 was utilized. The learning rate scheduler chosen was ReduceLROnPlateau to adapt to changes in the model's performance on the

validation set. Additionally, to prevent overfitting and save training time, an Early Stopping strategy was introduced, which terminates training prematurely if the performance metrics on the validation set do not improve for consecutive 2 epochs. The entire training process was conducted for a maximum of 1000 epochs to ensure the model had sufficient opportunities for convergence and optimization.

DFT Calculations: Vienna Ab initio Simulation Package (VASP) was employed to perform all density functional theory (DFT) calculations within the generalized gradient approximation (GGA) using the Perdew-Burke-Ernzerhof (PBE) formulation. The Gibbs free energy for each elementary step was calculated as: $G = E_{elec} + E_{ZPE} - TS$, where E_{ZPE} is the zero-point energy term, and T is the absolute temperature (here 298 K), S is the entropy. The cutoff energy of the plane wave set as 450 eV. The convergence criterion for total energy was set to 10^{-6} eV and atoms were relaxed until the Hellman-Feynman forces were less than 0.001 eV $\text{\AA}^{-1}$. The 15 Å vacuum layer was added to the surface to eliminate the artificial interactions between periodic images.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

All data supporting this study and the code are available on GitHub (<https://github.com/Zi-Jian-Gao/Exploring-structure-diversity-in-atomic-resolution-microscopy-with-graphs>).

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